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Bittensor Deregistration 101

A plain-English walkthrough of subnet deregistration, the on-chain mechanism that prunes inactive subnets and redirects emissions to the strongest.

▯ 2025-09-16 · filed by Shifa, Laron Campbell

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Bittensor is a decentralized machine learning network powered by its native token, TAO. At its core, the network is structured around its subnets, which are specialized communities where participants (miners and validators) collaborate and compete to produce digital projects and infrastructure. Each subnet operates as its own incentive mechanism, using the Yuma Consensus to allocate emissions (rewards in TAO) based on performance.

Subnet deregistration, reintroduced in August 2025 and deployed on September 16, 2025, is a mechanism designed to prune inactive subnets, ensuring the network remains competitive and focused on high-quality contributions. First deregistration is expected in the next week.

This announcement comes at the heels of observers realizing an astounding 30% of subnets are partially or completely inactive. With the cap for the ecosystem at the infamous 128 subnets, deregistration can be thought of as releasing the ecosystem's “dead weight”

In this new era of dTAO, only the strongest subnets survive, redirecting emissions and stake back to active ones.

Deregistration aims to protect the network by:

• Removing "dead weight" subnets that consume slots but contribute little.

• Refunding locked TAO to wallets, compensating alpha holders, miners, validators, and subnet owners fairly.

• Boosting incentives for surviving subnets as more TAO flows to root stake. Without deregistration, emissions (~7200 TAO daily) get spread too thin across 128 subnets, reducing incentives for innovation, stalling the exponential growth TAO has to offer.

When a new subnet owner applies for a slot, deregistration is automatically brought into play, triggering the removal of the weakest link. This process will in turn, foster a stronger, more competitive ecosystem.

Key Concepts to Understand Before Deregistration

• Immunity Period: New subnets get ~4-6 months (or ~120,000-180,000 blocks) of protection to build momentum without risk of removal.

New SNs are immune for the first 4 months, but SNs that have existed for 4+ months but aren't activated are the first candidates.

• NAV (Net Asset Value): Measures a subnet's health by comparing staked TAO to the market value of its alpha tokens (�).

NAV = (TAO staked in subnet) / (Market cap of � tokens in TAO equivalent).

If NAV > 1, the subnet is over-backed relative to its perceived value (undervalued or inactive).

• ADR (Alpha Distribution Ratio): The inverse of NAV (ADR = 1/NAV).

If ADR < 1 (equivalent to NAV > 1), the subnet is eligible for deregistration.

• EMA (Exponential Moving Average): A performance score based on a subnet's historical emissions share or activity. Low EMA indicates poor long-term contribution.

Deregistration targets SNs with the lowest EMA price.

• dTAO and Alpha Tokens

On deregistration, � dissolves back to TAO, often at a premium for holders if NAV > 1.

The Subnet Deregistration Process

Deregistration is automatic and on-chain, triggered by new subnet registrations rather than manual votes. Think of it as a market-driven “cleanup”, where only the strongest, thriving projects survive. Here's how it works step-by-step:

1. Eligibility Check:

• A subnet must be out of its immunity period (no protection for new builds).

• It must have NAV > 1 (or ADR < 1), meaning its staked TAO exceeds its � market value—indicating low demand or inactivity.

• Among eligible subnets, it must have the lowest EMA (poorest performance history).

• Subnets with active participants (e.g., high stake, regular weights set by validators) are naturally protected via higher EMA/ADR.

2. Trigger: New Subnet Registration:

• When someone registers a new subnet, the network checks for eligible candidates.

• If the total active subnets hit the 128 cap, the lowest-EMA eligible subnet is immediately deregistered to make room.

• Every new subnet potentially "kills" an old one. The first deregistrations were expected around September 23, 2025.

3. Refunds and Dissolution:

• Subnet Creator: The original TAO (from registration) is fully refunded to the creator's coldkey wallet. Under dTAO, any remaining lock costs are returned.

• Alpha Holders (dTAO Stakers): □ tokens are dissolved and converted to TAO at the NAV rate.

Example: If you hold □ worth 1 TAO market value but NAV=2, you receive 2 TAO—rewarding early or loyal holders in undervalued subnets.

• Staked TAO: All stake (from validators/miners) in the subnet is returned to owners' hotkeys/coldkeys. Delegated stake (e.g., from root validators) flows back to root.

• Emissions Impact: Dissolved TAO boosts root stake, funneling more emissions to surviving subnets. This can raise APY for dTAO holders in active subnets by 10-20% or more, depending on how many are removed.

4. Post-Deregistration Effects:

• The slot opens for the new subnet. • Network-wide emissions redistribute: Surviving subnets get a larger share of daily emissions.

What this means:

The best subnets will continue to thrive, attracting stake, setting competitive weights for the work on their subnet, and delivering consistent value to their alpha token holders.

As the number of inactive subnets are pruned and sheared, holders of alpha tokens can expect higher subnet quality over the next few months, with subnet owners working harder to communicate with updates, build to product, and provide as much value to their owners as possible, with risk of deregistration always looming over, driving innovation.

A possible benefit from deregistration could also be a stronger inflow from smaller projects to larger projects like @ridges_ai and @chutes_ai, with a consolidation in the space for the strongest projects to reign supreme.

Likewise for smaller or mid-level subnets project teams, like @polariscloudai or @metanova_labs, this consolidation can help to bring awareness to teams working hard and not seeing the upswing in their alpha

tokens just yet.

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Simplifying Bittensor subnet ecosystem through accessible, insightful articles. #TAO